Event Clips Overview

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Event Clips is a critical feature in GCXONE that automatically captures and retrieves short video segments around alarm events, providing operators with immediate context to verify and respond to security incidents. This feature ensures that operators have the exact video context needed without manually searching through hours of recorded footage.

Key Capabilities

Automatic Clip Capture

- Pre-Alarm Recording: Event Clips automatically captures video footage before an alarm triggers — typically 5 seconds before. This provides crucial context showing what happened leading up to the event.
- Post-Alarm Recording: The system continues recording after the alarm trigger — typically 5 seconds after — capturing the immediate aftermath and any ongoing activity.
- Configurable Buffer Times: The pre- and post-alarm buffer periods are configurable and can be adjusted based on specific requirements.

Supported Devices

Event Clips are automatically captured and retrieved for:

- ADPRO: Full support for event clip recording and retrieval
- Dahua: Automatic event clip capture and access
- Hikvision: Integrated event clip functionality
- Milestone: Event clip support through VMS integration

Device requirements:

- Device must support event-triggered recording
- Recording must be enabled on the device
- Device must maintain a recording buffer
- Network connectivity required for clip retrieval

Configurable Alarm Codes

Event Clips can be configured to capture clips for specific alarm codes:

- All Alarm Codes: Capture clips for all alarm types
- Specific Alarm Codes: Configure which alarm codes trigger clip capture
- Motion Detection: Capture clips for motion events
- Intrusion Detection: Capture clips for intrusion alarms

- Tamper Detection: Capture clips for tamper events

How It Works

Device-Side Recording

The device continuously records video to storage and maintains a rolling buffer for pre-alarm footage. When an alarm triggers, the device saves the pre-alarm buffer, continues recording for the post-alarm period, marks the clip with alarm event information, and makes it available for retrieval.

GCXONE Retrieval

When an alarm is received, GCXONE automatically requests the event clip from the device, retrieves it via network, processes it for immediate playback, and makes it available in the alarm interface.

Clip Duration

Standard configuration is typically 10 seconds total — 5 seconds before the alarm trigger and 5 seconds after. Custom configurations can be adjusted based on alarm type and severity, response time requirements, storage considerations, and network bandwidth.

Integration with Alarm Workflow

1. Device sends alarm to GCXONE
2. GCXONE automatically requests the associated clip
3. If AI is enabled, the clip is analyzed for human or vehicle detection
4. Alarm and clip are presented to the operator together
5. Operator views clip to verify the alarm
6. Operator makes decision and initiates appropriate response actions

Use Cases

Alarm Verification

- Quickly verify if an alarm is a genuine threat
- Immediately identify false alarms
- Understand what happened before and after the alarm

Incident Investigation

- First view of the incident scene
- Understand the sequence of events
- Collect and document video evidence

Integration with AI Analytics

- AI analyzes clips to classify alarms
- Bounding boxes show detected objects on clips
- Confidence scoring provides AI confidence levels
- False alarms filtered automatically before reaching operators

Configuration Options

Buffer Configuration

- Pre-Alarm Duration: Default 5 seconds before alarm
- Post-Alarm Duration: Default 5 seconds after alarm
- Total Clip Duration: Typically 10 seconds combined

Alarm Code Selection

- All Alarms: Capture clips for all alarm types
- Specific Codes: Select specific alarm codes for clip capture
- Priority-Based: Capture clips only for priority alarms

Storage and Retention

- Clips stored temporarily for operator access
- Retention period typically 30–90 days
- Automatic cleanup of old clips based on configured policies

Limitations and Considerations

Device Support

- Only supported devices can provide event clips
- Device must have recording enabled and maintain a recording buffer
- Requires network connectivity for retrieval

Storage Considerations

- Devices need sufficient storage for buffer
- High alarm rates generate many clips
- Balance retention period vs. storage capacity

Network Requirements

- Clips require bandwidth for retrieval
- Network must be reliable for clip access
- Multiple clips may be retrieved simultaneously

Best Practices

- Ensure at least 5 seconds of pre-alarm buffer for adequate context
- Include at least 5 seconds of post-alarm buffer
- Focus clip capture on critical alarm types and avoid capturing every minor event
- Regularly monitor device storage capacity and configure automatic cleanup
- Review and adjust alarm code selection periodically